

BEACONHOUSE NATIONAL UNIVERSITY



Lab 8 and Lab 10 Computer Networks

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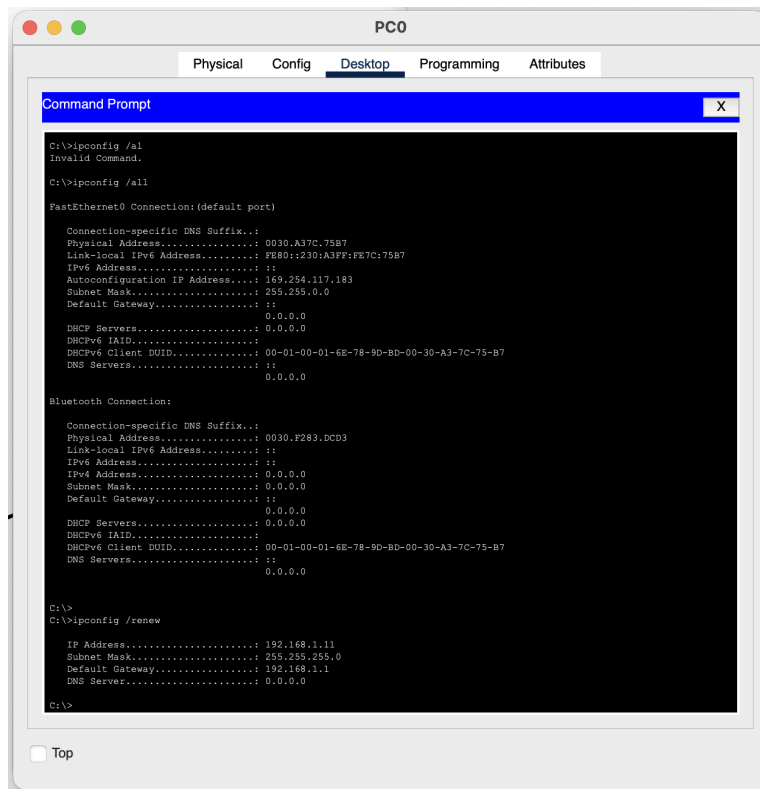
Due Date: 13 Mar, 2026

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Lab Instructor	Hasnain Ahmed
Semester	Spring 2026

Lab 8:

Task1-Task5:

ScreenShots:



Router>show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.1.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

Router>|

```

Router>enable
Password:
Router#show running-config
Building configuration...

Current configuration : 959 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$5.a6P4JqbNiMX0lusIfka/
!
!
ip dhcp excluded-address 192.168.1.1 192.168.1.10
!
ip dhcp pool pool1
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
!

```

PC0

Physical
Config
Desktop
Programming
Attributes

IP Configuration
X

Interface FastEthernet0

IP Configuration

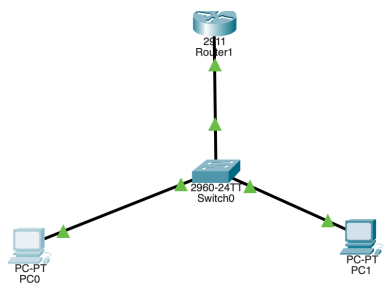
☒ DHCP
 ☐ Static

IPv4 Address 192.168.1.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0



Task6:

1. Write down the exact commands you used to set the hostname and passwords on HQ-Router:

```
Router> enable
Router# configure terminal
Router(config)# hostname HQ-Router
HQ-Router(config)# enable secret class
HQ-Router(config)# line console 0
HQ-Router(config-line)# password cisco
HQ-Router(config-line)# login
HQ-Router(config-line)# exit
```

2. Write down the exact commands you used to assign the IP address to GigabitEthernet0/0:

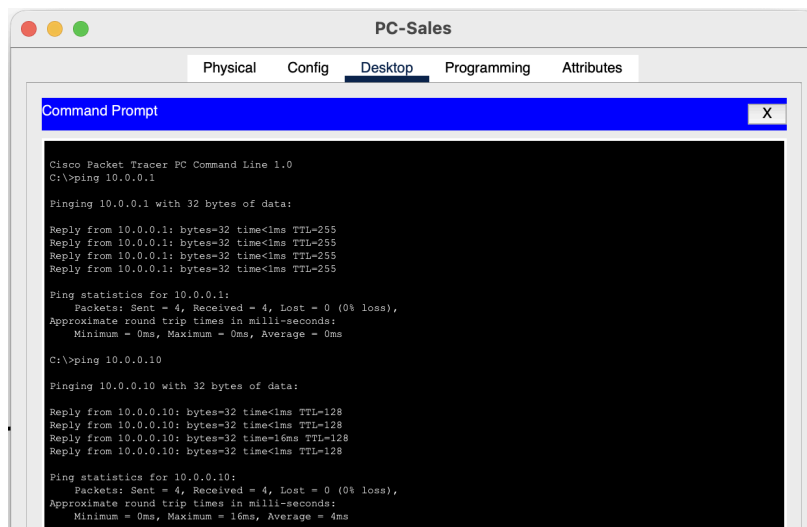
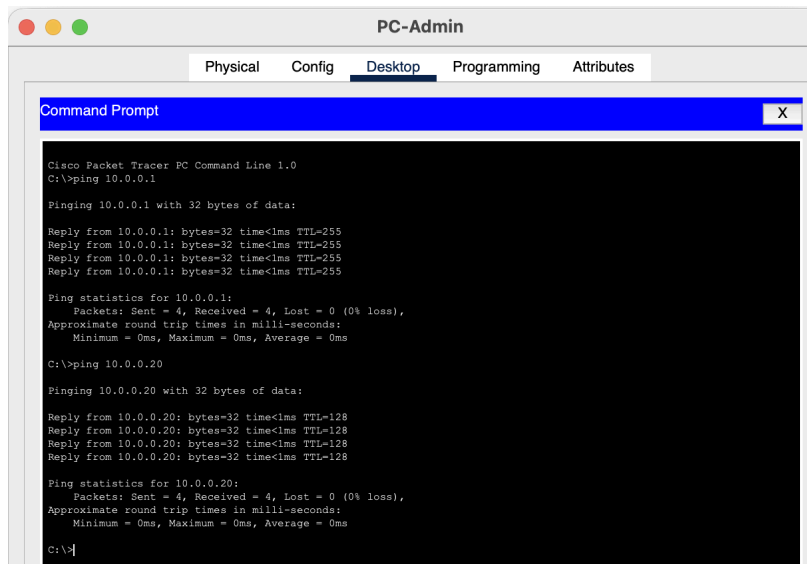
```
HQ-Router(config)# interface gig
HQ-Router(config)# interface GigabitEthernet0/0
HQ-Router(config-if)# ip address 10.0.0.1 255.255.255.0
HQ-Router(config-if)# no shutdown
HQ-Router(config-if)# exit
Write memory
```

3. Copy the output of 'show ip interface brief' from your router CLI here:

```
HQ-Router#show ip interface brief
Interface          IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0 10.0.0.1        YES manual  up                  up
GigabitEthernet0/1 unassigned      YES unset   administratively down down
GigabitEthernet0/2 unassigned      YES unset   administratively down down
Vlan1              unassigned      YES unset   administratively down down
HQ-Router#
```

4. Ping Test Results:

Ping Test	Packet Sent	Packet Received	Pass / Fail
PC-Admin pings Router (10.0.0.1)	4	4	Pass
PC-Sales pings Router (10.0.0.1)	4	4	Pass
PC-Admin pings PC-Sales (10.0.0.20)	4	4	Pass
PC-Sales pings PC-Admin (10.0.0.10)	4	4	Pass

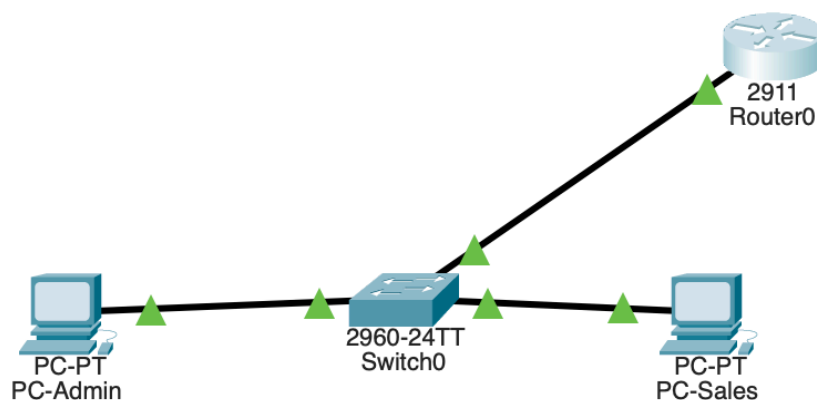


5. From 'show running-config', write the lines that confirm your hostname and passwords

```
HQ-Router#show running-config
Building configuration...

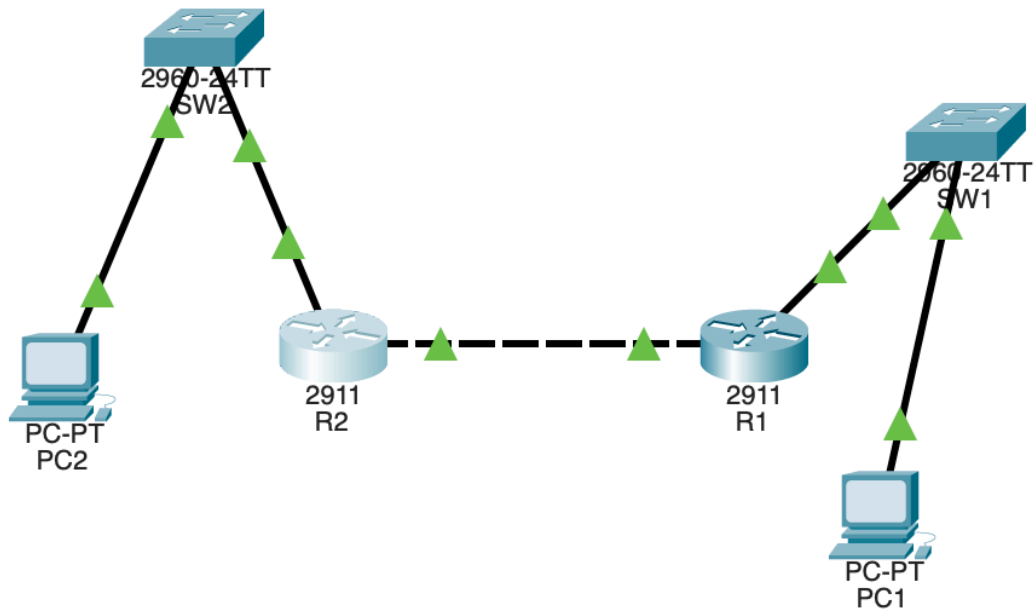
Current configuration : 784 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname HQ-Router
!
!
!
enable secret 5 $1$mERr$9cTjUIEqNGurQiFU.ZeCi1
!
!
!
!
!
ip cef
no ipv6 cef
```

6. Attach a screenshot of your completed Packet Tracer topology below:



Lab 10:

Topology ScreenShot:



'Show run | section dhcp' ScreenShot:


```

R2#show run | section dhcp
ip dhcp excluded-address 192.168.1.1 192.168.1.10
ip dhcp excluded-address 192.168.2.1 192.168.2.10
ip dhcp excluded-address 203.0.113.1
ip dhcp pool Pool1
  network 192.168.1.0 255.255.255.0
  default-router 192.168.1.1
  dns-server 8.8.8.8
  domain-name jeremysitlab.com
ip dhcp pool Pool2
  network 192.168.2.0 255.255.255.0
  default-router 192.168.2.1
  dns-server 8.8.8.8
  domain-name jeremysitlab.com
ip dhcp pool Pool3
  network 203.0.113.0 255.255.255.252
ip dhcp pool pool1
  default-router 192.168.1.1
R2#

```

'show ip dhcp binding' ScreenShot:

```

R2#show ip dhcp binding
IP address      Client-ID/      Lease expiration      Type
                Hardware address

```

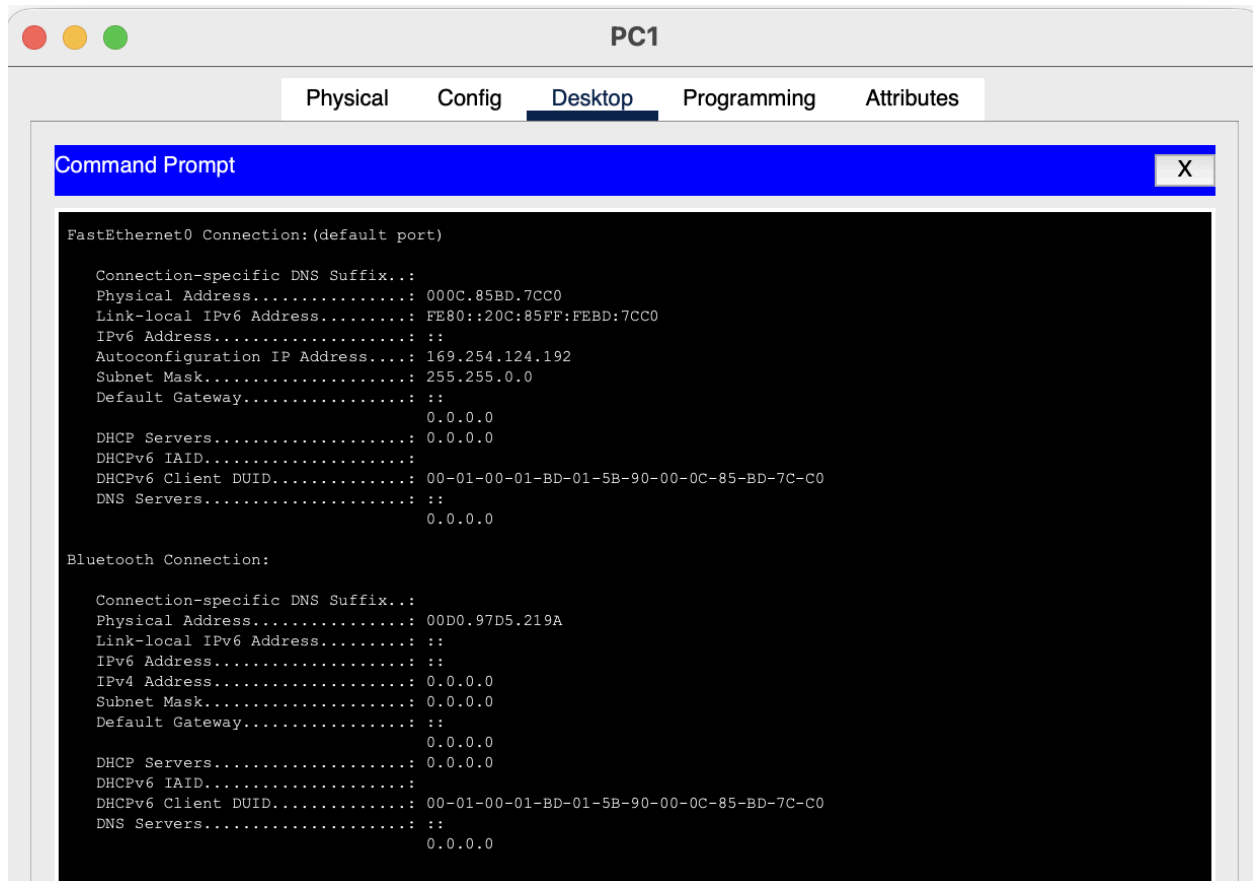
'show ip interface brief' ScreenShot:

```

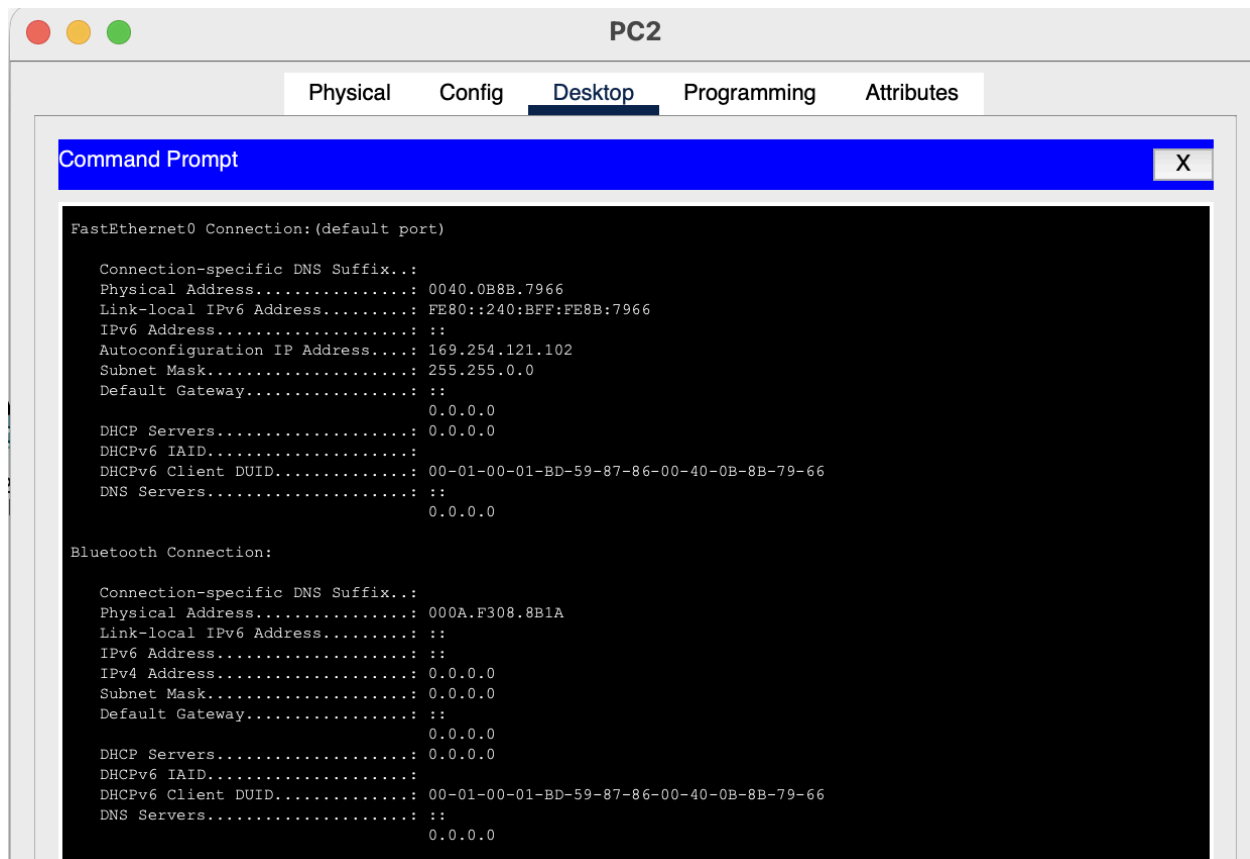
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  unassigned      YES DHCP    up          up
GigabitEthernet0/1  unassigned      YES unset   up          up
GigabitEthernet0/2  unassigned      YES unset   administratively down down
Vlan1          unassigned      YES unset   administratively down down

```

ipconfig /all from PC1 ScreenShot:



ipconfig /all from PC2 ScreenShot:



Completed Answer Questions:

Q1)After running show ip dhcp binding, what IP address did PC2 receive?

Answer:

The assignment specifies that addresses .1 through .10 must be excluded for Pool2. Since DHCP servers typically assign the first available address in the pool, and the first ten are reserved, PC2 will receive the eleventh address in that subnet.

Q2)What IP address did R1's G0/0 interface receive from R2's DHCP Pool3? Why is this the only possible IP it could receive?

Answer:

203.0.113.2

Q3)In your own words, explain why 'ip helper-address' is needed in this topology. What problem does it solve?

Answer:

The ip helper-address is necessary because PC1 and the DHCP server (R2) are located on different subnets separated by a router (R1).

Q4)Record the IP addresses received by each device below

Device	IP Address Received	Default Gateway	Pool Used
PC1	192.168.1.11	192.168.1.1	Pool1
PC2	192.168.2.11	192.168.2.1	Pool2